CLASS 11-CBSE MATHEMATICS

CHAPTER: LIMITS AND DERIVATIVES-SET-1

TIME : 1Hr **Mx.Marks:30**

SECTION A

Q.I Answer the following questions **[1X3 = 3 MARKS]**

1. If $f(x) = x + 2$, then $f'(f(x))$ at $x = 4$ is
2. $\lim_{x\to\infty} \frac{(3x-1)(2x+5)}{(x-3)(3x+7)}$ is equal to
3. If $y = \frac{(1-x)^2}{x^2}$, then $\frac{dy}{dx}$ is

SECTION B

Q.II Answer the following questions **[2X5 = 10 MARKS]**

4. If $f(x)$ has a derivative at $x = a$, then $\lim_{x\to a} \frac{xf(a) - af(x)}{x - a}$ is equal to
5. Find $\lim_{x\to 3} f(x)$, where $f(x) = \{4,\ if\ x > 3\ \ x + 1,\ if\ x < 3$
6. $If\ y = \sqrt{\frac{x}{a}} + \sqrt{\frac{a}{x}},\ prove\ that\ 2xy\frac{dy}{dx} = \left(\frac{x}{a} - \frac{a}{x}\right)$
7. If for f (x) = $\lambda x^2 + \mu x + 12$, f' (4) = 15 and f' (2) = 11, then find λ and μ.
8. Differentiate each of the following from first principles:
   (i) e^{-x} (ii) e^{3x} (iii) e^{ax+b} (iv) $x e^x$

SECTION C

Q.III Answer the following questions **[3X3 = 9 MARKS]**

9. Evaluate $\lim_{x\to 2} \left(\frac{1}{x-2} - \frac{2}{x^2-2x}\right)$
10. Evaluate $\lim_{x\to 0} \frac{\sin x^2 \left(1-\cos x^2\right)}{x^6}$

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11. Differentiate $\frac{ax^2+bx+c}{px^2+qx+r}$

SECTION D

Q.IV Answer the following questions **[4X2 = 8 MARKS]**

12. Differentiate in two ways, using product rule and otherwise, the function

(1 + 2 tan x) (5 + 4 cos x). Verify that the answers are the same.

13. Evaluate the following

a) $\lim_{x\to 1}\left(\frac{1}{x-1}-\frac{2}{x^2-1}\right)$

b) $\lim_{x\to 3}\left(x^2-9\right)\left[\frac{1}{x+3}+\frac{1}{x-3}\right]$

c) $\lim_{x\to 1}\frac{x^4-3x^3+2}{x^3-5x^2+3x+1}$

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